First-flush likelihood under a common-start, continuous evolutionary process

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Abstract

We are close to being able to expand beyond our own planet, and we see nobody else doing it. The hard-steps model explains the silence by making civilisations rare, but Mills et al. (2025) argue that Earth's record gives poor support for improbable steps. If lineages usually get there in the end, the older ones had a large head start, and some should now be visible. We work out how large that head start may be, in flat space with perfect detection of expanded galaxies. A clock tied to each planet's birth must keep loud lineages extremely rare. A clock that starts from a shared cosmic event and carries progress between planets allows a lead of about 14 million years. Once lineage durations vary independently, the sky constrains our rank rather than the lead: we sit in a thin first flush of emergences arriving around now, in our galaxy and in others. Scored by full non-indexical conditioning, this clock produces observers like us 10^7 to 10^9 times as often as a hard-steps lottery if expansion is loud, and ties with it if expansion is filtered, which puts a price on the loudness premise.

1. The observation

Start with something we can see. Humanity is one technological civilisation, arriving at cosmic age $t_0 = 13.8$ Gyr, in a sky with no sign of a civilisation that has spread across its galaxy. Any account of where civilisations come from has to produce that sky, seen from here, now. We have not yet expanded; treating today as the point where we could is an approximation that holds while the remaining delay is short against the Myr scales below.

The hard-steps account produces it through rarity. A lineage must pass n transitions whose typical waiting times exceed the life of its planet, so the chance of emerging within time t grows as t^n and almost all planets never get there (Carter 1983; Watson 2008; Snyder-Beattie et al. 2021). Hanson et al. (2021) add expansion: the rare lineages that emerge and spread visibly ("grabby" civilisations) occupy volume where later ones can no longer appear independently, so anyone who finds themselves in an empty sky arrived before the local deadline.

Mills et al. (2025) pull in a different direction. Working through the usual candidate steps, they argue that happening once on Earth is weak evidence of improbability. The first lineage to cross a threshold tends to close the niche behind it, and several thresholds coincide with planetary changes (rising oxygen is the standard example) that arrived late for reasons having nothing to do with biological luck. On their reading, Earth's record looks like a sequence of doors opening.

Earth's record, though, is one planet. Suppose the doors do open in sequence and most suitable lineages walk through them eventually. Older planets had more time: in the Galactic model of Lineweaver, Fenner, and Gibson (2004), about 75% of stars in the Galactic habitable zone are older than the Sun. Hanson posed the resulting problem in 2010, for life that spreads easily

between worlds and then evolves easily on them (Hanson 2010), and returned to it after Mills et al. (Hanson 2024). It is the question this note takes up: how much sooner can older lineages have emerged, given that we see none of them?

The clock we test runs as follows. Progress accumulates in lineages that can move between environments, so a planet's birth starts nothing. All lineages draw on progress from a shared cosmic starting point (our candidate is the first appearance of carbon chemistry) and take a characteristic total time from there. Pieces of this picture are old: Hanson (1998) allowed one difficult beginning followed by steady advance; galaxy-wide synchronisation has its own literature (Annis 1999; Ćirković and Vukotić 2008); Sharov and Gordon (2013) argued for a lineage older than Earth. Ponfilenok (2026) works the closest related model: a common start, synchrony arising from many small steps, and silence bounding the spread of completion times inside one galaxy. Here the bound comes from expansion fronts between galaxies, and what is added is the comparison of clock laws, galaxy saturation, start ranks and independent duration tails under that bound, and a likelihood comparison with hard steps.

We offer no posterior odds on the process clock against hard steps. The calculation assumes the process clock and finds what it costs; Section 6 compares likelihoods at matched sky boundaries, without priors. The accounts differ in where they put the improbability: inside the transitions, or in the timing.

2. Setup

Count suitable spacetime in Earth-equivalent units. A **zone** here is a run of conditions in which evolution can proceed and retain its gains. (As illustration only: liquid, an energy gradient and churn, on an icy body as readily as on a planet, with the arena a lineage needs growing as it complexifies.) Let N be the effective number of retained-progress trajectories still in play in a galaxy today. We take $N = 10^{10}$ as the central value, normalised for scale as 10^{11} stars times 0.1 effective trajectories per star, and run the calculation across 10^8 to 10^{11} where it matters. That normalisation is a Fermi scale, not a count of Earth-like planets. The counted trajectories $N(t)$ grow from near zero at cosmic age 0.2 Gyr, tracking the growth of effective Goldilocks capacity as enrichment spreads. They are not planets being born. A counted trajectory is assumed to stay in play; dropping out is left out of this calculation.

Count one lineage per effective trajectory. Two clocks (Figure 1):

- *Planet clock.* For this comparison only, identify each trajectory with one planet. A lineage on a planet born at time b becomes able to begin loud expansion — crosses the loud threshold — at $b+T$. We call that crossing **emergence**; it names an operational endpoint, not a biological stage.
- *Inherited clock.* Progress runs from a shared start s. A trajectory already in play when the running total reaches T crosses at $s+T$. A trajectory entering the count at $b < s+T$ joins the frontier wherever it has got to and also crosses at $s+T$. One entering at $b > s+T$ is credited with the crossed state on entry, giving crossing time $\max(b,\, s+T)$. Read that formula as bookkeeping for a shared frontier. An arena-resolved version would follow a trajectory between sites as the arena it needs grows; the one clock compresses the whole history into T.

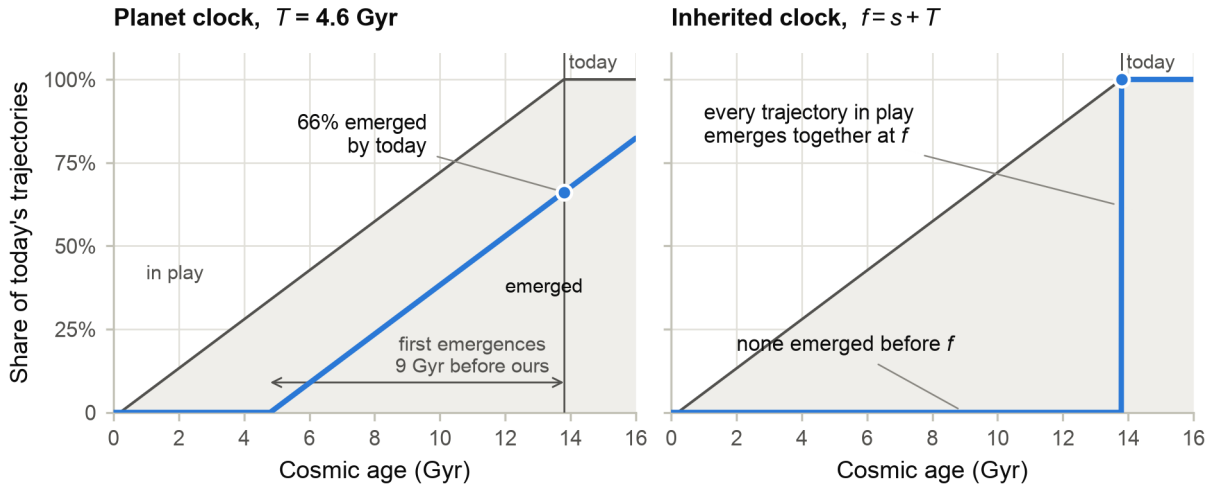


Figure 1: Two clocks, same trajectories. Grey: share of today's effective trajectories in play at each cosmic age (a linear ramp from 0.2 Gyr). Blue: share that has crossed the loud threshold. Left, the planet clock at $T = 4.6$ Gyr: crossing trails a planet's birth by T, so two thirds of trajectories have crossed by today, the first 9 Gyr before us. Right, the inherited clock: nothing crosses before the shared date $f = s + T$, when every trajectory in play by then crosses at once in this one-clock bookkeeping. Drawn with $f = t_0$, so $\delta = 0$.

A lineage that has crossed turns loud (spreads through its galaxy and keeps spreading) with probability q. Loud expansion is taken from Hanson et al. (2021) and Olson and Ord (2021), as a front at $v = 0.5c$ that visibly alters every galaxy it crosses. Whether anyone expands like that is itself an assumption, and Section 6 puts a price on it. It becomes visible to us later still, after the front has crossed its galaxy and the light has travelled. We mostly need one combination:

$$\Lambda = qN,$$

the mean number of lineages per galaxy that cross the loud threshold and turn loud, if every counted trajectory crosses. Call it the loud yield. With m crossed lineages in a galaxy, the chance that at least one is loud is $1-(1-q)^m$, close to $1-e^{-qm}$ whenever q is small. Λ governs most of what follows; q and N stay separate in the appendix and where the difference matters.

The geometry follows Olson (2018) and Olson and Ord (2021), simplified. Space is flat. Galaxies are scattered independently at an average density of 0.01 per cubic megaparsec: pick any volume and the galaxy count averages $0.01\times$ volume, with Poisson scatter around it. Two ways for a predecessor to have shown itself:

- *We could see it.* Once a front has crossed its home galaxy (30 kpc across, about 0.2 Myr at $0.5c$), the galaxy counts as visible, and its light needs r/c to reach us from distance r.
- *It could be here.* The front itself reaches us after r/v.

For each condition separately, count the galaxies expected to satisfy it and call the count μ. Independent galaxies give an empty-sky probability of $e^{-\mu}$. Our working line is $\mu = -\log 0.05 \approx 3$: beyond it, an empty sky has under a 5% chance *inside this model*. The inputs are assumed, several guessed to an order of magnitude, so each bound below is an edge for those inputs. Giving a bound a probability in the real world would take priors over the inputs and a rule for sampling observers.

Detection here is perfect: every crossed galaxy registers. Existing waste-heat searches look for galaxies re-radiating a large share of their starlight in the mid-infrared, in samples of limited

size and depth (Wright et al. 2014; Griffith et al. 2015). Our μ counts loud galaxies present in the volume. Turning it into a survey probability would need the intercepted starlight fraction, distances, dust and catalogue completeness. Appendix A collects the equations.

3. If every planet starts its own clock

This section imposes the opposite rule from the one we favour, to see what it costs. Read each counted trajectory as one planet, starting from nothing at that planet's birth and needing the same time T. Earth caps T at 4.6 Gyr for any history run entirely on one planet, and 4.6 Gyr is the choice friendliest to this branch, since a shorter T moves every other planet's emergence earlier as well.

On the same ramp, counted planets have been forming since 0.2 Gyr — earlier than rocky planets were plausibly common. That choice is not generous: early planets lengthen the head start and tighten the ceiling below. Starting the ramp at 2 or 4 Gyr instead, with today's count fixed, loosens it to 4.4×10^{-8} or 1.3×10^{-7}. At $T = 4.6$ Gyr every planet born before cosmic age 9.2 Gyr has emerged: two thirds of today's stock, with the first emergences 9 Gyr before ours. That count keeps every planet suitable until emergence. There is no destruction, no stellar-lifetime cut and no closing habitable window. A cut would remove some old planets before they could emerge and soften the ceiling below, while leaving a spread of billions of years among the survivors. Holding these emergences to an empty visible sky requires $\Lambda \lesssim 2.1 \times 10^{-8}$. In words: fewer than one galaxy in fifty million may produce a loud lineage, ever, out of its 10^{10} counted trajectories ($q \lesssim 2.1 \times 10^{-18}$ at the central N). Counting only planets old enough to have emerged by today, about one galaxy in 73 million hosts an expander now. The arrival condition gives a ceiling eight times weaker, the volume factor at $0.5c$.

Trust the shape of this result more than its digits (Figure 2). It integrates over gigaparsecs and most of cosmic history, where flat space and a fixed galaxy density are crude. The shape says: planet clocks plus common emergence leave only one place to hide, inside q. Almost every emerged lineage has to stay quiet, die young, or expand in a way nobody can see. Something has to filter them.

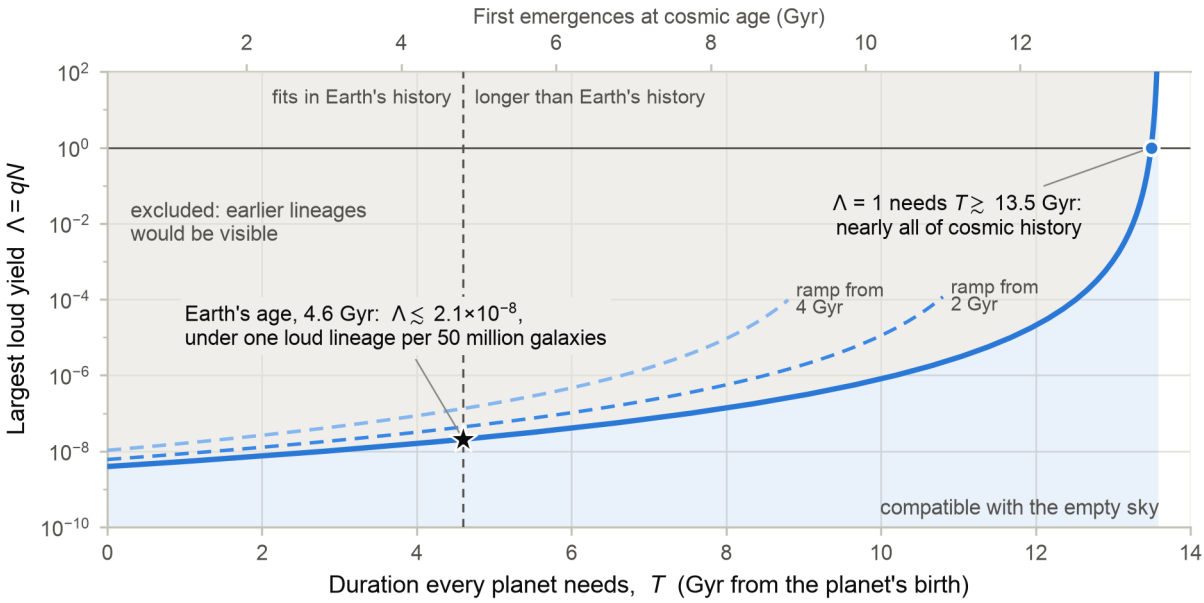


Figure 2: Planet clocks. The largest loud yield $\Lambda = qN$ compatible with an empty sky ($P(\text{empty}) \geq 5\%$, visibility at $v = 0.5c$), against the duration T every planet needs; the top axis gives the cosmic age of the first emergences, $0.2\,\text{Gyr} + T$. Grey: excluded. At Earth's age, 4.6 Gyr, the ceiling is $\Lambda \approx 2.1 \times 10^{-8}$. Yields of order unity need durations approaching the age of the universe. Dashed: the rare-limit ceiling (Appendix A) with the planet ramp starting at 2 or 4 Gyr instead, today's count fixed; later starts shorten the head start and loosen the ceiling. The solid curve below $T \approx 1.2$ Gyr also uses the rare limit.

4. A shared starting gun

Now let progress carry between planets, running from the shared start s. Planet age stops mattering: what matters is the date $f = s + T$ at which the process first emerges anywhere, and how far that date sits ahead of ours. Write the lead as $\delta = t_0 - f$.

Take the extreme case first: every lineage shares the same start and duration, so one frontier date f describes all of them. This case cannot describe our own position (under it we emerge at f too, and $\delta = 0$). Its caps count other galaxies only: every lineage in our galaxy shares our start and duration and crosses with us, so none is ahead by construction. Any spread within a galaxy puts some of them ahead, and Section 5 takes that up. It is the limit the spread-out versions approach, and it shows the geometry cleanly.

For $\Lambda \gtrsim 10$ the visibility condition gives $\delta \lesssim 13.7$ Myr (Figure 3). Arrival allows 27 Myr; tightening the line to $\mu = 1$ gives 9.6 Myr. At $\Lambda = 1$ and 0.01 the leads are 16 and 63 Myr. The plateau is a volume statement: a lead δ exposes loud galaxies out to about $c\delta$, and at 13.7 Myr that sphere reaches ~ 4 Mpc and holds about three galaxies. Once those galaxies are loud, more lineages per galaxy change nothing (a galaxy cannot be louder than loud; Figure 8). Cosmic expansion is negligible over 4 Mpc, which is why this number is firmer than Section 3's.

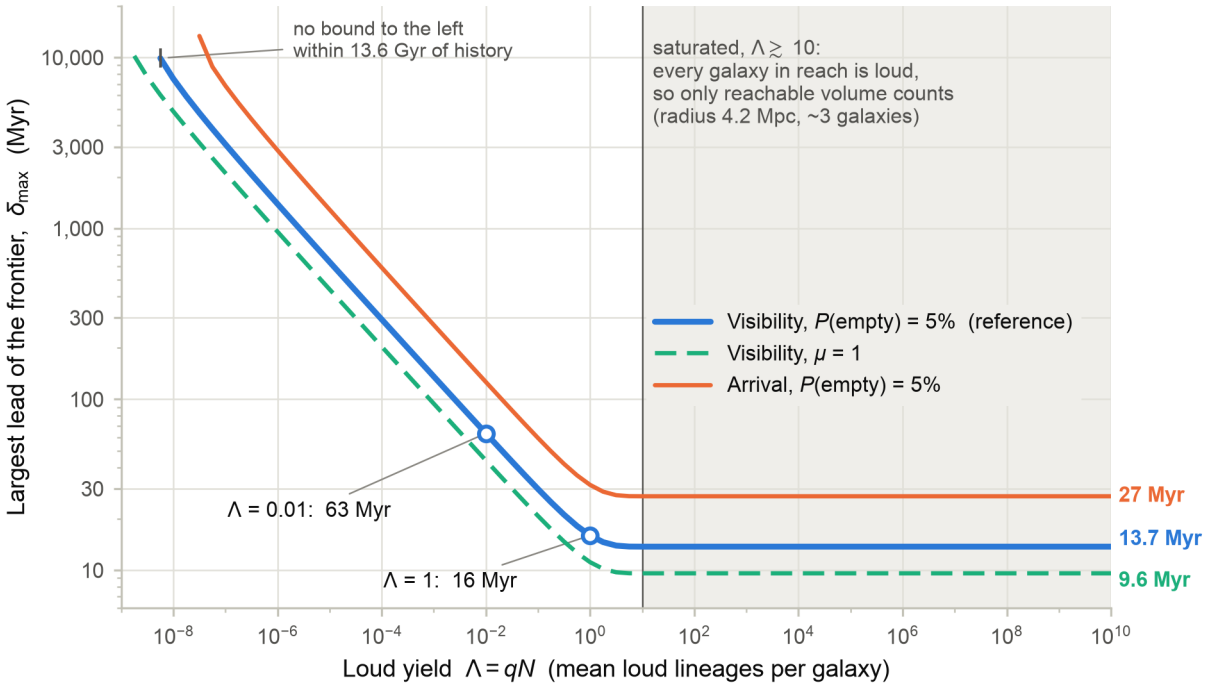


Figure 3: The head-start cap. Largest lead δ of a shared emergence frontier over our own emergence, against loud yield Λ, for three sky conditions. In the grey band ($\Lambda \gtrsim 10$) every galaxy in reach is loud and the cap is set by volume alone: 13.7 Myr for visibility at $P(\text{empty}) = 5\%$, 9.6 Myr at $\mu = 1$, 27 Myr for physical arrival. Left of the tick, no lead within the 13.6 Gyr history is excluded.

Real starts will scatter. Let the start of each galaxy's clock fall somewhere in a window of width W, all durations still equal, expansion saturated (Λ large). Emergence order then copies starting order. We emerged today; the galaxy that started first emerged δ before us, where δ is exactly our own distance into the window. The sky caps δ, and the cap translates into a position inside the window (Figure 4):

Window W	Largest δ	Our start must lie within
100 Myr	32 Myr	first 32%
200 Myr	38 Myr	first 19%
500 Myr	47 Myr	first 9.5%
5 Gyr	84 Myr	first 1.7%

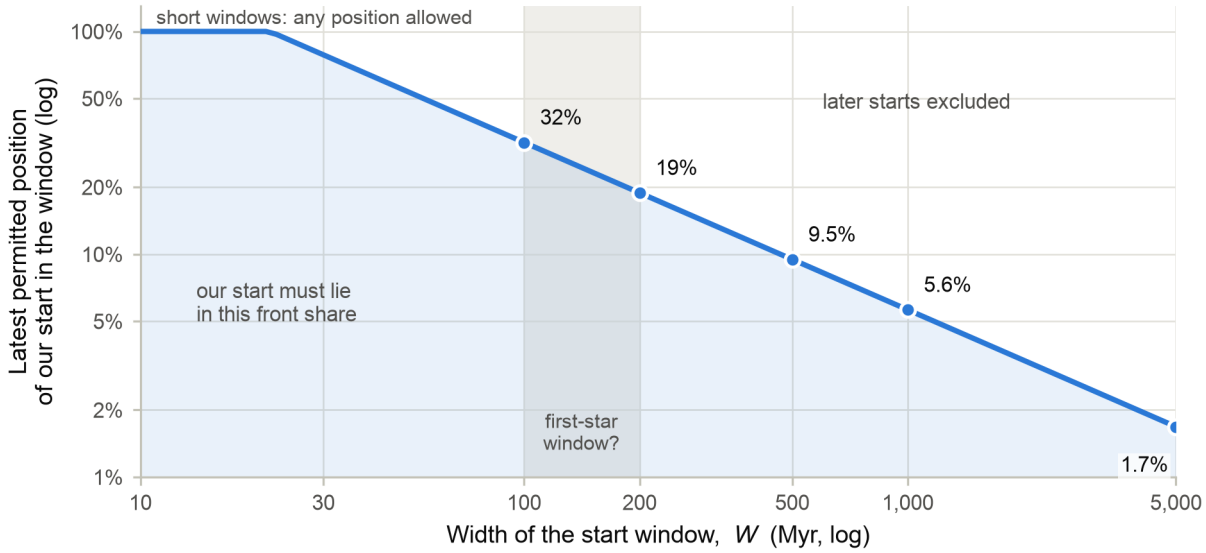


Figure 4: Spreading the start. Latest permitted position of our start inside a uniform start window of width W, with saturated yield and visibility at $P(\text{empty}) = 5\%$. Below about 20 Myr any position is allowed; past that, widening the window shrinks the permitted share almost as fast. Grey band: an illustrative 100–200 Myr window for first-star enrichment. Each point is an edge in the model, not a probability.

Read the right column as an edge, in the sense of Section 2: start later than it and an empty sky drops below a 5% chance in the model. It is not a probability that we really started inside the edge; that would need priors and an observer rule. Widen the window and the bill arrives as a position nearer its front. Fix our start at the middle of a 5 Gyr window instead, and the bill arrives as $q \lesssim 1.4 \times 10^{-16}$, back in filter territory.

What could fire starting guns across many galaxies within a few hundred Myr? It must reach everywhere and switch on conditions for accumulation. An earlier universal switch exists on paper: for roughly 7 Myr, when the universe was 10–17 Myr old, the cosmic microwave background alone kept surfaces between freezing and boiling (Loeb 2014). That Goldilocks interval was probably wasted. Temperature was right before the ingredients were: in the standard picture almost no star-forming haloes, heavy elements or rocky planets yet existed to use the warmth. The first stars and their carbon are the first gun with material to work on (Bromm 2013; Stacy and Bromm 2014). The sky calculation cannot select it: sliding the start date and adjusting T leaves all emergence dates unchanged. Halo assembly and mixing will widen whatever window the first stars set, so the 100–200 Myr rows are a case to test against enrichment histories.

Picture one way the clock could run; nothing in the calculation depends on it. Early stages (chemistry, protocells) ask little of their surroundings: liquid water and an energy gradient, available on icy moons and small bodies as well as planets, and far more numerous than Earth-like zones. Progress there is set by the frontier, the best stage reached anywhere within reach, not by the age of any one body. Later stages need bountiful Goldilocks environments like Earth's surface, so a planet that opened early gains little from waiting. Moving between sites requires transfer and establishment, whose rates may be low (Adams and Napier 2022); rock exchange inside a birth cluster could correlate sibling systems (Belbruno et al. 2012).

Whatever the mechanism, the model's own reading of Earth is short. Earth's LUCA, around 4.2 Gyr ago (Moody et al. 2024) and shortly after Earth formed, dates when our lineage reached Earth, not its origin. Earth's record is the last third of the clock, run at the same rate as the first two thirds: our lineage arrived near the front and held its lead there, and the planet added

no speed of its own. Planet age drops out too. The older stars of Section 1 give no head start, and under the spreads of Section 5 nearly every trajectory trails ours, the typical one by about a Gyr at the 1% row, since the sky puts us in the fast tail.

From a start at cosmic age 0.2 Gyr, our elapsed duration is 13.6 Gyr: the start choice read back to us. The next section moves the *typical* duration off that value.

5. Variation in duration

So far every lineage in a galaxy has shared one start and one duration. Split the two:

$$C_{i,g} = S_g + D_i,$$

with S_g a start shared across galaxy g and D_i the duration of lineage i. Section 4 priced scatter in S_g. To isolate duration scatter below, set $S_g = S$: one cosmic starting gun for every galaxy. A galaxy goes loud through its earliest loud lineage. How much earlier that happens depends on how independently its lineages develop. Shared ancestry and exchange of material can correlate their progress. The estimates below take durations as independent between zones, so each zone is a separate chance to emerge early. That is the severe end for precedence and the case the process clock must survive.

One choice drives everything in this section: the left tail of the duration distribution. The boundary percentile sits near 10^{-12} — about 6.8σ out on a normal curve — where no physical argument supplied here constrains the shape. Distributions that look alike near their centre can differ by orders of magnitude out there. A normal tail thins forever and always permits a few much faster lineages. A duration law with a hard physical floor $D_{\min}$ permits none before $S + D_{\min}$; if that floor date falls close to today and probability accumulates slowly just above it, as it would for many sequential stages, emergences can bunch near the frontier even with independent durations. A floor alone does not create that bunching (the density just above it matters), but it removes the long early tail that drives the extreme ranks below. Read the normal rows as arithmetic for one tail shape, not as a measurement of duration spread in general.

Give each lineage an independent duration, normally distributed with spread σ_T. Hair (2011) spread emergence times normally for up to 10^4 civilisations and stressed the first one's long head start, placing the centre of his distribution near today. Invert it: the first emergences arrive around now, so where does the centre sit? Hair's placement is excluded at home before the sky is consulted: with the mean at today, half our galaxy's zones have emerged, and home is loud unless $\Lambda \lesssim 1$, a filter again. Counting other galaxies alone, it would still need $\sigma_T \lesssim 2$ Myr, since the fastest of 10^{10} draws lands about $6.4\sigma_T$ ahead of the mean against the 14 Myr plateau as a rough allowance: durations equal to one part in 10^4. Local chemistry, waits for planetary conditions and sterilising accidents (Gowanlock, Patton, and McConnell 2011) will not hold that line over 13.6 Gyr.

Take a spread of hundreds of Myr and solve for the earliest mean the sky allows, building each galaxy's loud probability from its 10^{10} draws. That is a boundary, not a fit: placing the mean later only makes an empty sky likelier, so the sky alone gives inequalities. The rows below give the most we can have emerged by today, and the earliest the typical zone can emerge. Then check home. The setup so far counted other galaxies; inside the Milky Way the expected loud lineages ahead of us are $\Lambda F(t_0)$, where F is our emergence percentile, and our chance of being first loud at home is $e^{-\Lambda F(t_0)}$. (Fronts cross the galaxy in under a million years, so travel time barely moves these rows.) At $q = 1$:

Duration spread σ_T	Share of zones emerged by today, $F(t_0)$, at most	Expected loud lineages ahead of us in the Milky Way	Chance home is still quiet
30 Myr	5.9×10^{-10}	5.9	0.003
136 Myr (1% of 13.6 Gyr)	6.2×10^{-12}	0.06	0.94
408 Myr (3%)	2.7×10^{-13}	0.003	1.00

The 30 Myr row passes the external sky and fails at home. Our own existence supplies what the sky lacks: an observer rate that falls as the mean moves later. Scored by full non-indexical conditioning (Section 6), the likelihood of the mean placement peaks near $\mu_{\text{vis}} \approx 1$, 22 Myr later than the 1% boundary and 61 Myr later than the 3% one; at both, the boundary rows sit within a factor of 3 of the peak. At 30 Myr the peak sits 11 Myr later with 0.6 loud lineages ahead at home, and the boundary row is 300 times less likely.

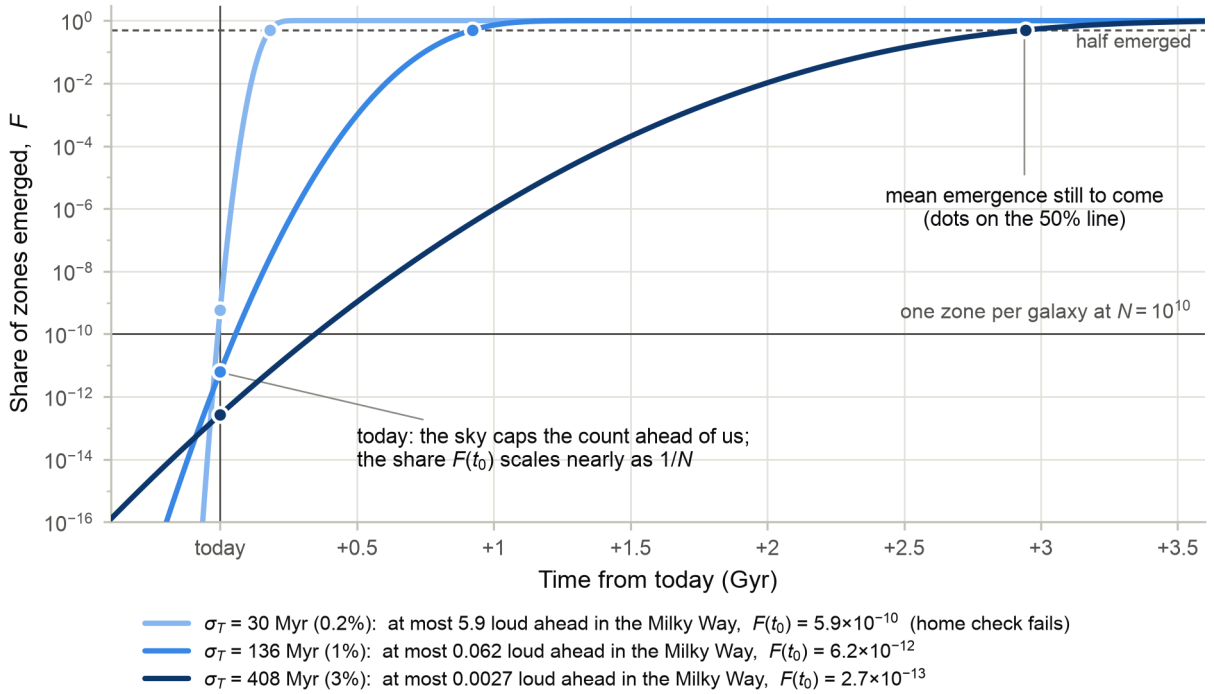


Figure 5: The first flush. Share of zones emerged, F, against time from today, for three spreads of independent normal durations at $q = 1$, each with its mean at the earliest the external sky allows, $P(\text{empty}) = 5\%$. Dots at today give $F(t_0)$; dots on the 50% line give the mean emergence, 0.18, 0.92 and 2.9 Gyr from now. The grey horizontal line marks one zone in a 10^{10}-zone galaxy. These are boundary rows; the likelihood peak under full non-indexical conditioning sits 11–61 Myr later, too close to show at this scale. The 30 Myr row fails the home check.

These rows assume independent normal durations (Appendix A; Figure 5); tail shape and correlations between lineages change them. Correlation leaves the expected count $\Lambda F(t_0)$ unchanged at a fixed marginal percentile, but changes the probability that no lineage has yet turned loud. Two points. First, the sky fixes how many lineages came before us, not what fraction of zones we lead. Rerunning the 1% row with N from 10^6 to 10^{11} moves $F(t_0)$ from 3.4×10^{-8} to 7.0×10^{-13}, close to $F \propto N^{-0.94}$, while the expected count ahead of us at home, $NF(t_0)$, stays between 0.03 and 0.07 and the typical zone stays 0.73–0.96 Gyr in the future (Figure 6). The percentile is mostly a restatement of N. The two accounts couple N and F differently. Under hard steps F is

the lottery's success share, fixed per zone, and the sky caps NF. Under the process clock the sky holds NF near 0.06 and F follows N. Fewer effective trajectories, through shared ancestry or a smaller count, make our position less extreme here and leave the hard-steps lottery untouched. Second, one tail position covers home and away. At spreads of 100 Myr and up, external galaxies bind harder than ours — many galaxies, each with 10^{10} draws, the nearer ones dominating μ by count — and a position deep enough for them leaves 0.06 expected loud lineages ahead of us at home. At 30 Myr the home condition binds instead, at $F(t_0) \sim 10^{-10}$. Either way the picture is a first flush: a thin leading edge arriving around now in galaxy after galaxy, the bulk behind it. Eventual emergence can still be the rule; the sky charges for it in percentile rather than in q. The flush also has a near date. On the 1% boundary row, the median wait for the first external front is about 34 Myr, given that none has arrived and none is visible today, and 45 Myr at the likelihood peak. In both cases we would see some neighbouring galaxies turn loud before their fronts arrived. On these rows an empty sky is a brief condition.

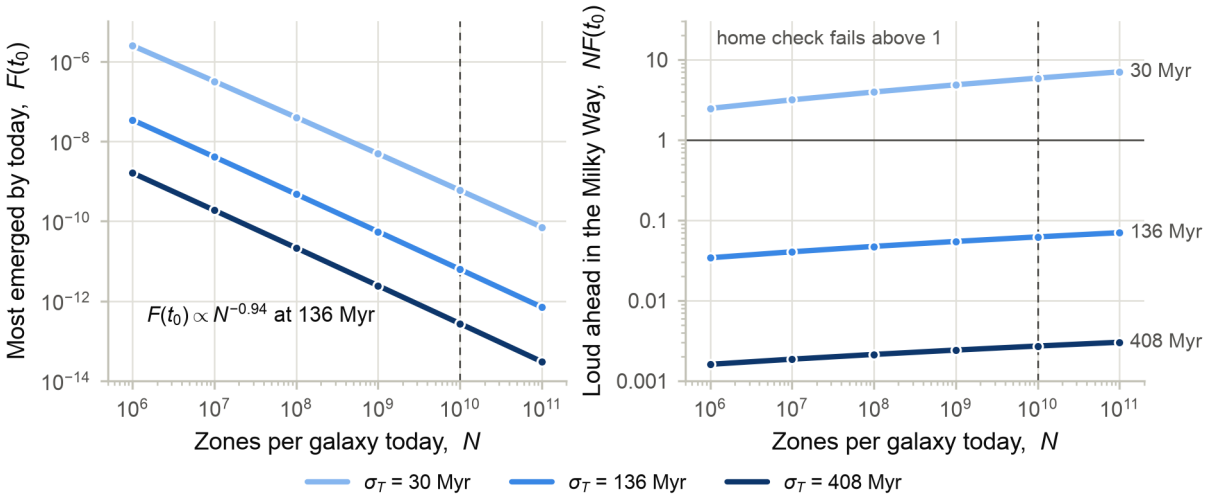


Figure 6: The sky fixes the count, not the share. Left: the most that can have emerged by today, $F(t_0)$, against zones per galaxy N, at the 5% sky boundary with $q = 1$. Right: the same rows as expected loud lineages ahead of us in the Milky Way, $NF(t_0)$. Over five decades of N the share falls almost as $1/N$ while the count moves by about a factor of two. Dashed: the central $N = 10^{10}$.

$F(t_0)$ also gives the cleanest comparison between the two accounts. Both deliver a tiny emerged share today, and both put our luck somewhere. Under hard steps, F is a success share within planetary lifetimes: almost every zone fails. For a single same-aged cohort its cumulative chance scales as t^n; on a cosmic clock at 13.8 Gyr with $n = 5$, one more Gyr multiplies F by $(14.8/13.8)^5 \approx 1.42$. That is a scale comparison, not a demographic forecast over zones of mixed ages. Under the process clock, the eventual emerged share of suitable zones is close to one and $F(t_0)$ is small only because today sits on the leading edge. At the 1% spread it climbs from 6.2×10^{-12} to one half in 0.92 Gyr, a factor of 8×10^{10}. One lottery decides whether a zone ever gets there within its planet's lifetime; the other decides where a zone sits in a race that almost every suitable zone eventually finishes. Under these two appearance laws their futures differ. Hard steps keep independent emergences sparse over the next Gyr, even though an approaching grabby front remains possible. The process clock sends an emergence wave through existing zones over the next few Gyr unless earlier arrivals occupy them first. A rising rate of nearby independent emergences is the distinguishing signature, not contact by itself.

6. Scoring the clocks

To score the accounts against each other we use full non-indexical conditioning (Neal 2006): a hypothesis is rated by the probability that some observer, somewhere, has our evidence, here an emergence near t_0 under an empty sky in a quiet home galaxy. While such observers are rare, that probability is proportional to their expected rate. Other sampling rules give other answers (Bostrom 2002; Olson and Ord 2021); this one counts observers and needs no reference class. Hold every branch on the same external boundary, so the 5% sky factor cancels, and take the emergence rate per galaxy at t_0 times the chance that home is still quiet. The empty sky can be paid for in two places: before emergence, keeping emergence rare and every emerged lineage loud ($q = 1$); or after it, letting emergence be common and setting q at its sky ceiling, a late filter. Crossing those with three clocks gives the rate of observers like us, per galaxy per Gyr at t_0, counting only those whose home is still quiet:

Clock	Improbability before emergence	Late filter
Hard-steps lottery, $n = 3$–9	1×10^{-9} – 2×10^{-7}	$\leq 7 \times 10^{8}$
Planet clock, $T = 4.6$ Gyr	1.5×10^{-9}	7×10^{8}
Process clock, $\sigma_T = 136$ Myr	3	3×10^{8} – 3×10^{10}

The lottery puts each planet's emergence at age a with density $\propto a^{n-1}$, with and without a 5 Gyr habitable window, its rate solved for the sky; Hanson et al. (2021) put n near 3 to 9. Its late-filter cell is a bound: with q free the lottery can be made easy, and no planet emerges more than once, so the rate cannot pass the planet birth rate. The process clock's late-filter range runs from a mean completion $3\sigma_T$ after today to one at today; with q free, nothing in the sky picks between them.

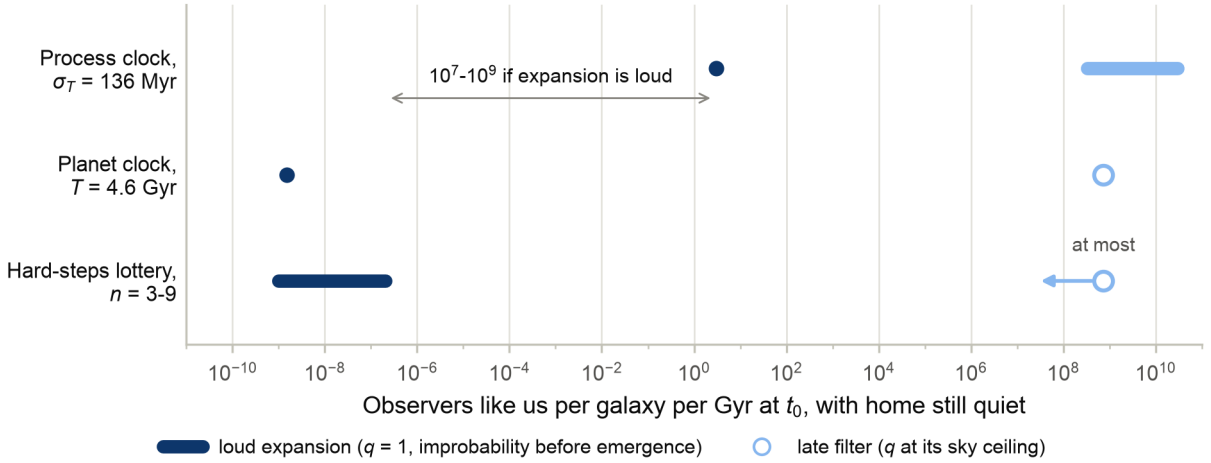


Figure 7: Observers like us under full non-indexical conditioning, per galaxy per Gyr at t_0 with home still quiet, every branch on the same 5% sky boundary. Dark: loud expansion, with the improbability before emergence. Light: a late filter, with q at its sky ceiling. Bars are ranges: over $n = 3$–9 with and without a 5 Gyr habitable window for the lottery, and over the mean placement for the process clock's late filter. The lottery's late-filter value is a bound, since no planet emerges more than once.

Read down the left column first (Figure 7). If loud expansion is real, the process clock beats the

lottery by 10^7 to 3×10^9. A steep emergence wave has only tens of Myr of past light cone to keep quiet; emergences spread over Gyr must be thin everywhere to stay unseen. The lottery comes closest at $n = 9$ with no habitable window, where its emergences crowd towards the present. Read the right column next. If expansion is filtered, the clocks tie to within a factor of 40, since a sky with nothing loud in it constrains no clock. The rule leans towards the late filter in every row, as observer-counting rules do (Grace 2010); that lean is the objection "why assume anyone expands visibly?", given a price. To prefer loud expansion, prior odds on it must beat about 10^9 under the process clock and 10^{15}–10^{18} under the lottery. The late filter stays live until a first loud galaxy is seen, and on the process clock with loud expansion that sighting is near: neighbours turn loud before their fronts arrive. Within the process clock at $q = 1$, a 30 Myr spread scores about the same as the 1% row, its extra emergences offset by a home galaxy that is rarely still quiet; a 3% spread scores 0.016. Each branch here is tuned to its boundary: the process clock through its mean placement, the lottery through its rate. The likelihood is about 20 Myr wide in the mean at the 1% spread and about one e-fold wide in the lottery's rate, so flat priors over a Gyr or more of placement, or over many decades of rate, cost each branch 10^2–10^3 and leave the ordering intact. These are likelihood ratios at one exclusion boundary for each branch, not posteriors: they carry no priors over q, σ_T or N, and they inherit the flat geometry.

7. Conclusions

The process clock itself is the largest thing left out: we assume a characteristic duration and inherited progress, and derive neither from chemistry. Within Section 5, the unidentified left tail of the duration law is the main modelling choice; a physical minimum duration would change the percentiles and arrival times qualitatively. Also assumed: a lineage that turns loud stays recognisable once it has crossed its galaxy; counted trajectories are never destroyed or dropped; each either runs the full course or contributes nothing.

For the 14 Myr cap only the nearest few megaparsecs enter, and there flat, unclustered galaxies are a fair first pass — though the actual Local Group arrangement would shift the cap by factors of order unity. Section 3 uses the same flat machinery over cosmological distances and should be read as a scaling statement until redone in an expanding universe with an evolving galaxy population. Clustering also correlates neighbours' enrichment histories and hence their start times.

Our galaxy enters Section 5 through the expected count of earlier loud lineages and the probability of none under independent durations. A full treatment needs positions within the galaxy, and priors over q, σ_T and N to turn the boundary likelihood ratios of Section 6 into posteriors. They are deferred, with any attempt to derive durations from accumulated opportunity.

To summarise the account: planet-bound clocks, with each planet running its own abiogenesis and long evolution from scratch, make the sky empty by making loud lineages vanishingly rare per galaxy. That is the pressure that pushes the explanation toward intrinsic hard steps. A shared cosmic start with inherited progress makes the sky empty a different way. With a single shared emergence date and appreciable loud yields, the permitted lead is about 14 Myr. Spreading the starts permits a longer lead, but still places us in the first flush of arrivals on cosmological timescales. With independent, normally distributed durations and $q = 1$, the examples allow between about 6 and 0.003 earlier loud lineages at home and place us among the first 10^{-10} to 10^{-13} of zones at $N = 10^{10}$, a share that moves almost inversely with N, with the mean emergence still to come. Correlated histories can change probabilities at a fixed marginal percentile. In the independent normal-duration model with a 1% spread and $q = 1$, the boundary timing leaves at most an expected 0.06 earlier loud lineages in the Milky Way, giving at least a 94% chance of none, and the likelihood peak sits 22 Myr later. On that branch the first flush is local as well as global, and the first external front is tens of Myr away. Nothing

here shows the process clock is true. It shows the clock can be run without a filter, and states the timing it then requires. Under full non-indexical conditioning, and with loud expansion, that timing produces observers like us 10^7 to 10^9 times as often as a hard-steps lottery. Without loud expansion the two tie. The process clock makes the loudness premise about 10^9 to afford, against 10^{15}–10^{18} for the lottery.

What would test this? Not new civilisations. On the 1% row the Milky Way produces about three independent emergences per Gyr, and the first external front is tens of Myr away, so the chance of seeing either within a century is below one in a million. The process clock makes nearer predictions about simpler life. If the early stages run as a shared frontier through small wet bodies, simple life should be common, and related to ours: life on Europa or Enceladus would share deep biochemistry with Earth's (chirality, core metabolism), where an Earth origin or a hard early step predicts barren moons or unrelated chemistry. Mars is a weaker test, since rock exchange with Earth could relate the two without any frontier. Further out, the accounts disagree about complex life. Under the process clock nearly every suitable trajectory finishes the course, the typical one about a Gyr behind us, so biospheres as developed as Earth's, with the oxygenated atmospheres that preceded complex life here, should be common on Earth-like planets. Under the hard-steps lottery most such planets fail at one of the steps and stay barren or microbial. Ocean-moon missions and the direct-imaging surveys of nearby temperate planets planned for the 2030s and 2040s can begin on both.

Data and code availability

Configurations, saved outputs and verification files sit in the accompanying repository. `make baseline`, `make inference` and `make start-spread` regenerate the sweep numbers; `uv run python -m lineage_cap.iid_tail` regenerates the normal-tail and first-arrival table at `results/iid_tail/normal_tail_estimate.csv`; `MPLCONFIGDIR=.mplconfig uv run python figures/make_note.py` regenerates the figures. `make check` runs the tests.

This note, its figures and data are licensed under CC BY 4.0; the code is MIT licensed. An interactive version of the calculation is at https://www.fergusfettes.com/lineage-cap/.

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Appendix A. Calculation

Times are cosmic ages in Gyr; distances are Mpc; $c = 306.6014$ Mpc Gyr^{-1}. Effective trajectory counts grow as

$$N(t) = N_0 \,\mathrm{clip}\left(\frac{t-s}{t_0-s}, 0, 1\right)^{\alpha},$$

with $s = 0.2$ Gyr, $\alpha = 1$, $N_0 = 10^{10}$ at the centre of a 10^8–10^{11} sweep. Emerged counts by deadline t under the two clocks:

$$m_{\mathrm{planet}}(t;T) = N(t-T), \qquad m_{\mathrm{inherited}}(t;f) = \mathbf{1}_{t\geq f} N(t),$$

where $f = s + T = t_0 - \delta$. A galaxy with m emerged lineages hosts a loud one with probability

$$p_g(t) = 1 - (1-q)^{m(t)},$$

evaluated as $-\mathrm{expm1}[m \log 1p(-q)]$. Visibility and arrival impose deadlines

$$t_{\mathrm{vis}}(r) = t_0 - D/v - r/c, \qquad t_{\mathrm{arr}}(r) = t_0 - r/v,$$

with $D = 0.03$ Mpc and launch delay set to zero. Writing a deadline as $t_k = t_0 - d_k - r/c_k$,

$$\mu_k = 4\pi n_g \int_0^{R_k} r^2 p_g[t_k(r)]\, dr, \qquad P(\mathrm{empty}) = e^{-\mu_k}.$$

The working line is $\mu_k = -\log(0.05)$. When every galaxy in reach is loud, $p_g = 1$ inside the integral and

$$\delta_{\max} = d_k + \left(\frac{3\mu_k}{4\pi n_g c_k^3}\right)^{1/3},$$

which is the 13.7 Myr plateau (Figure 8): $d_{\mathrm{vis}} = D/v \approx 0.2$ Myr plus the light-travel radius that encloses an average of three galaxies.

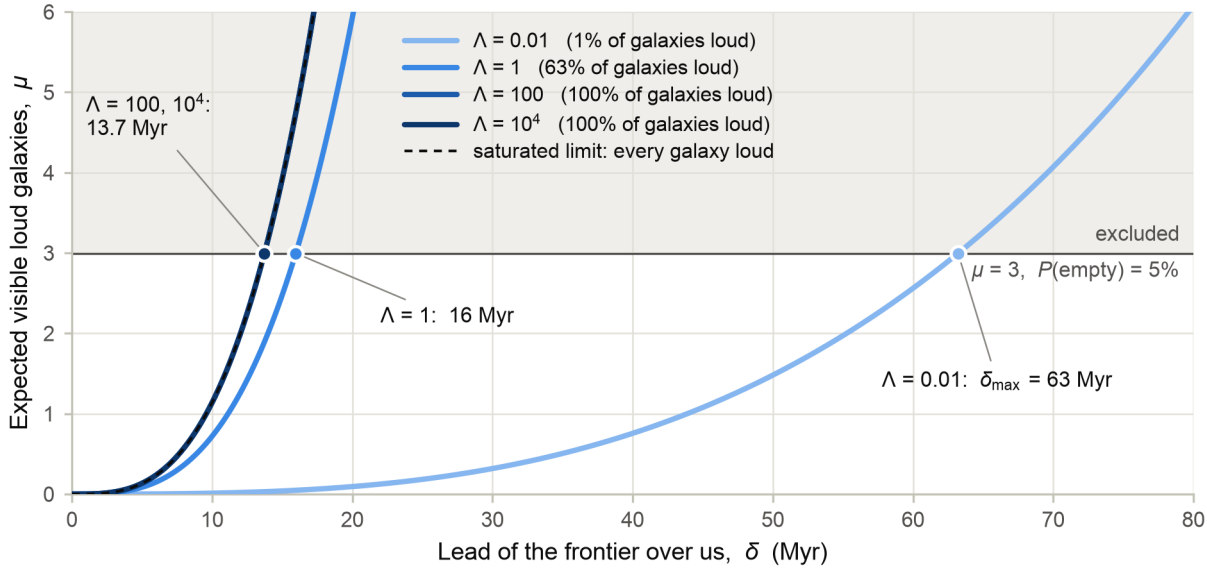


Figure 8: Why the cap flattens. Expected visible loud galaxies μ against lead δ at four loud yields, darker for larger Λ. Dashed: the saturated limit with every galaxy loud, $\mu = \frac{4\pi}{3} n_g [c(\delta - D/v)]^3$. At $\Lambda = 100$ and 10^4 the model curves lie on it, so extra lineages change nothing. Each $\delta_{\max}$ is where a curve crosses $\mu = 3$; grey is excluded.

For starts spread uniformly over width W with one draw $S = s_{\text{first}} + U$ per galaxy and a common duration, the share of galaxies emerged by t is

$$F(t) = \text{clip}\left(\frac{t - t_0 + \delta}{W}, 0, 1\right),$$

and averaging over the start draw comes before the at-least-one step: $p_g = F[1 - (1-q)^{N(t)}]$. With H independent start groups per galaxy, $p_g = 1 - \{1 - F[1 - (1-q)^{N(t)/H}]\}^H$.

The per-lineage estimates in Section 5 use a single shared start S for all galaxies and durations $D \sim \mathcal{N}(m, \sigma_T^2)$ drawn independently per zone. Writing F_D for the duration CDF, the emerged share by t is $F(t) = F_D(t - S)$ and a galaxy is loud by deadline t with probability

$$p_g(t) = 1 - [1 - qF(t)]^{N(t)}.$$

Inserting this in the visibility integral and solving $\mu_{\text{vis}} = -\log(0.05)$ for m at fixed σ_T gives the emerged shares $F(t_0)$ tabulated in Section 5 ($q = 1$, $N(t_0) = 10^{10}$). The Milky Way column is $qN(t_0)F(t_0)$, the mean count of loud lineages by t_0 in one galaxy; the probability that a named lineage emerging at t_0 is the first loud one in its galaxy is $\exp[-qN(t_0)F(t_0)]$ to Poisson accuracy, with travel-time corrections across the galaxy below 1% at these spreads. For first arrival, the same p_g is integrated with the physical-arrival deadline: a galaxy at r must have emerged by $t_0 + \Delta - r/v$. Galaxies form a Poisson process marked by their first loud date, so conditioning on no arrival and no visible loud galaxy at t_0 leaves a Poisson process on the galaxies excluded by neither: at radius r, those loud after $\max(t_0 - D/v - r/c,\ t_0 - r/v)$. Solving for the Δ at which their expected arrival count reaches $\log 2$ gives the median wait quoted in Section 5: 33.5 Myr at $\sigma_T = 136$ Myr, against 19.6 Myr conditioning on no arrival alone, and 45 Myr at the likelihood peak. Quadrature over lookback time with the normal CDF reproduces the table; values are saved in `results/iid_tail/normal_tail_estimate.csv`. Tail shape matters at the order-of-magnitude level and these rows should be read as normal-tail arithmetic.

For the N dependence, every trajectory count is multiplied by a constant and the boundary re-solved. Across $N(t_0) = 10^6$ to 10^{11}, $F(t_0) \propto N^{-k}$ with $k = 0.91$, 0.94 and 0.95 at $\sigma_T = 30$, 136 and 408 Myr; z_0 moves from 5.40 to 7.08 at 136 Myr. Values are in `results/iid_tail/n_sweep.csv`.

The likelihood peak maximises $\log f_D(t_0 - S) - \mu_{\text{vis}} - qN(t_0)F(t_0)$ over the mean at fixed σ_T: observer rate, empty external sky and quiet home. Values are in `results/iid_tail/fnc_peak.csv`. For the planetary ramp, the rare-limit ceiling is $\Lambda_{\max} \simeq 3L\mu_*/[\pi n_g c^3 (L - T - D/v)^4]$ with $L = t_0 - s$, which gives the 2 and 4 Gyr start values quoted in Section 3.

The likelihood comparison counts emergences at t_0 per galaxy. For the process clock that is $N(t_0) f_D(t_0 - S)$, with f_D the normal density, times $\exp[-qN(t_0)F(t_0)]$ for a quiet home. For the planet clock on the linear ramp the birth rate is N_0/L with $L = t_0 - s$. Putting the improbability before emergence gives a rate $\Lambda_{\max}/L$, with $\Lambda_{\max} = 3L\mu_*/[\pi n_g c^3 (L - T - D/v)^4] = 2.06 \times 10^{-8}$ at $T = 4.6$ Gyr; putting it after emergence, at $q = \Lambda_{\max}/N_0$, gives N_0/L. The lottery gives a planet of age $a < W$ an emergence rate kna^{n-1}, so a galaxy's rate is $(N_0/L)\, k \min(t - s, W)^n$; with $W = \infty$ the rare-limit boundary has the closed form $k = \mu_*(n+1)/[4\pi n_g c^3 (N_0/L) X^{n+4} B(3, n+2)]$, $X = t_0 - D/v - s$, which the solver reproduces. The process clock's late filter takes the largest q that keeps $\mu_{\text{vis}} = \mu_*$ at a fixed z_0, with rate $N(t_0) f_D$ and quiet home $\exp[-qN(t_0)F(t_0)]$. `uv run python -m lineage_cap.observer_rates` writes these tables; the rates are in `results/iid_tail/observer_rates.csv`.

Sweep integrals use Gauss–Legendre quadrature with a change of variables that resolves the thin partly-loud shell at the edge of the visible volume; inversions use bisection. Doubling the quadrature order shifts saved boundaries by at most 1.4×10^{-7} (500 baseline boundaries) and

1.1×10^{-7} (600 start-window boundaries, 60 probability ceilings) in relative terms. Those figures describe the arithmetic. The scenario choices are uncertain by far more.